



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM - 638 401

Module Test - II - May - 2026

II Semester

22CH203 & ENGINEERING CHEMISTRY II

1	(i)	Two reactions are compared where one has a low peak but small energy difference, and another has a high peak but large energy difference. Both occur under identical conditions. Scientists evaluate speed and stability. Integrate the concepts of activation energy and energy difference to determine which reaction is faster and which is more stable. (3 Marks - [Ap/C, 2])
2	(ii)	A reaction shows a very high peak followed by a steep drop in energy. The process releases large energy quickly. Safety concerns arise in handling this reaction. Engineers evaluate risks. Assess the reaction suitability considering both safety and feasibility. (2 Marks - [Ap/C, 2])
3	(i)	A reaction occurs in a single liquid phase where catalyst and reactants are fully miscible. The system shows rapid reaction rates and formation of a temporary species before final products are obtained. Infer the role of phase uniformity and intermediate formation in determining the reaction efficiency and pathway. (3 Marks - [An/C, 2])
4	(ii)	A homogeneous reaction shows that increasing catalyst concentration beyond a point does not significantly increase the reaction rate. Interpret the kinetic limitation observed in the system. (2 Marks - [An/C, 2])
5	(i)	A factory compares two processes where one has low activation energy but low energy output, and another has high activation energy with high energy output. The aim is continuous production. Engineers assess efficiency. Examine the trade-offs and prioritize the more efficient process for large-scale production. (3 Marks - [An/C, 2])
6	(ii)	A student draws an energy profile where reactants are higher in energy than products with a single peak. The graph represents energy versus reaction coordinate. The student must interpret reaction nature. Infer the reaction type and justify using the energy profile characteristics. (2 Marks - [An/C, 2])

7	(i)	A homogeneous catalytic reaction exhibits dependency on solvent polarity and catalyst concentration. Changing the solvent alters the reaction rate and product distribution. Analyze how solvent properties and catalyst concentration collectively influence the reaction kinetics and selectivity. (3 Marks - [An/C, 2])
8	(ii)	A solution-based catalytic reaction produces high yields but requires additional steps to recover the catalyst after completion. Determine the trade-off between efficiency and practicality in this catalytic system. (2 Marks - [Ap/C, 2])
9	(i)	In a laboratory, a reversible reaction is monitored where both forward and reverse reactions occur. The forward reaction requires less energy compared to the reverse process. Scientists plot an energy diagram and observe different activation energies for both directions, along with distinct energy levels for reactants and products. Determine how the difference in forward and reverse activation energies governs the directionality and equilibrium behavior of the reaction. Infer the implications for reaction feasibility and stability. (5 Marks - [Ap/C, 2])
10	(i)	A research team studies two catalysts for the same reaction: one finely powdered and another in large solid chunks. Both are chemically identical but show different efficiencies under identical conditions. The powdered catalyst performs significantly better in terms of conversion rate. Interpret the observed difference in catalytic performance by integrating concepts of surface phenomena and reaction mechanisms. Formulate the relationship between catalyst structure and efficiency. (5 Marks - [An/C, 2])
11	(i)	A chemical plant studies a reaction where reactants must cross an energy barrier before forming products. Engineers observe that even though the reaction is thermodynamically favorable, it proceeds slowly at room temperature. They analyze the energy profile diagram and identify a peak representing a high-energy intermediate state. Additional heating significantly increases the reaction rate. Analyze the role of the energy barrier in controlling the reaction progression and determine how the identified intermediate state influences the reaction rate under varying temperature conditions. Justify the observed rate enhancement with temperature. (5 Marks - [An/C, 2])
12	(i)	A chemical plant operates a reaction where gaseous reactants pass over a solid catalyst bed. The engineers observe that increasing the surface roughness of the catalyst improves the reaction rate significantly. However, beyond a certain point, further increase in temperature reduces efficiency due to unwanted side effects. The process involves adsorption of reactants, surface reaction, and product release. Analyze the sequence of steps occurring in this catalytic system and determine how surface properties and temperature collectively influence the overall reaction efficiency. Justify the dominant stage controlling the rate under the given conditions. (5 Marks - [An/C, 2])

13	(i)	In a pharmaceutical reaction, a specific enzyme catalyzes only one type of substrate while ignoring structurally similar molecules. The reaction proceeds rapidly in small quantities of the enzyme, and repeated cycles show no loss in activity. The system reaches the same equilibrium position as before catalysis. Interpret the behavior of the enzyme in terms of specificity, efficiency, and its influence on equilibrium, and justify how these factors contribute to industrial applicability. (5 Marks - [An/C, 2])
14	(i)	A chemical manufacturing unit observes that adding a small quantity of a substance increases the reaction rate significantly, while another similar-looking additive slows it down. The process engineers also note that the rate enhancement does not change the final equilibrium composition. They aim to classify these effects properly and optimize the process conditions. The system is sensitive to impurities, and small variations lead to large performance differences. Analyze the observed behavior and determine the classification of the substances affecting the reaction rate, and justify how their roles influence reaction pathways and equilibrium without altering final composition. (5 Marks - [An/C, 2])
15	(i)	A chemical plant is producing ammonia using a reaction that proceeds very slowly under standard conditions. Engineers introduce an iron-based catalyst into the system and observe that the reaction rate increases significantly without increasing temperature or pressure. The catalyst remains unchanged after multiple cycles. Energy profile analysis shows a reduced peak for the catalyzed pathway. Analyze how the introduction of the catalyst modifies the reaction pathway and justify the observed increase in reaction rate using energy considerations and catalyst properties. (5 Marks - [An/C, 2])
16	(i)	An industrial plant compares two catalytic systems: one where reactants and catalyst are in the same phase, and another where the catalyst is solid while reactants are gases. The first system shows uniform reaction but is difficult to separate, while the second system shows surface-based reactions with easier separation. The engineers must select the best system for large-scale production. Examine both catalytic systems and determine their comparative advantages and limitations in terms of mechanism, efficiency, and industrial feasibility, leading to a justified selection. (5 Marks - [An/C, 2])
17	(i)	A physicist calculates the mass of a nucleus and compares it with the sum of individual nucleon masses, finding a discrepancy. This difference is associated with energy release during nucleus formation. Determine the significance of this mass discrepancy and relate it to nuclear binding energy. (3 Marks - [U/C, 2])
18	(ii)	Two atoms have the same atomic number but different atomic masses, yet they react identically in chemical reactions. Justify why their chemical behavior remains unchanged despite mass differences. (2 Marks - [U/P, 1])

19	(i)	A graph plotted between number of nuclei and time shows a continuously decreasing curve. The decrease is rapid initially and becomes slower later. The curve never reaches zero even after long durations. Scientists study this graphical trend to understand decay behavior. The aim is to interpret its meaning Interpret the graphical representation of radioactive decay and determine what it indicates about the variation in decay rate. (3 Marks - [Ap/P, 2])
20	(ii)	A scientist studies the average lifetime of radioactive nuclei in a sample. The decay constant is known and used to interpret the duration of existence of nuclei. The scientist wants to relate average lifetime with half-life. This helps in understanding decay duration more clearly. The analysis is conceptual. Analyze the relationship between mean life and half-life and determine how they are connected through decay behavior. (2 Marks - [U/C, 1])
21	(i)	A laboratory detects two isotopes of the same element where one is used in medical diagnostics due to its emission properties, while the other is non-reactive. Both have identical electron configurations but differ in nuclear composition. Infer how nuclear differences enable one isotope to be suitable for diagnostic applications while the other remains inactive. (3 Marks - [Ap/C, 2])
22	(ii)	A nuclear model shows that increasing neutron number beyond a limit leads to instability in certain elements. Interpret why excessive neutrons reduce nuclear stability. (2 Marks - [U/C, 1])
23	(i)	A medical physicist observes radioactive tracers under different environmental conditions such as temperature and pressure. The decay rate remains unchanged in all cases. Repeated experiments confirm identical results. The physicist evaluates whether external factors influence decay. The goal is to ensure reliability in applications. Analyze the influence of environmental conditions on radioactive decay and justify the observed consistency. (3 Marks - [U/C, 2])
24	(ii)	A radioactive sample reduces to one-fourth of its original activity over a period of time. The researcher observes that the reduction follows a consistent halving pattern. This helps in tracking decay progression. The focus is on interpreting the number of decay stages. The observation is based on fractional decrease. Determine how many decay stages are involved when activity reduces to one-fourth of its initial value. (2 Marks - [U/P, 1])
25	(i)	A chemical plant compares two isotopes of chlorine used in parallel reactions and observes identical reaction pathways but slight differences in reaction rates. The engineers suspect that while electron configurations remain unchanged, atomic mass may influence certain measurable parameters. They aim to evaluate how isotopic substitution impacts both chemical and physical behavior in controlled conditions. The system operates under constant temperature and pressure with identical reactant concentrations. Analyze the observed similarities and differences in the behavior of the isotopes and determine how atomic structure and mass contribute to these variations under the given controlled conditions. (5 Marks - [U/C, 3])

26	(i)	A nuclear experiment tracks a reaction where a target nucleus is bombarded by particles, resulting in a different element being formed. The reaction obeys strict conservation of mass number and atomic number. Scientists calculate the energy released using measured mass differences. The reaction is being optimized for isotope production. Formulate the governing principles that validate the reaction representation and infer how energy changes are evaluated using measurable quantities. (5 Marks - [U/C, 3])
27	(i)	In a laboratory experiment, two isotopes of hydrogen are allowed to diffuse through a porous membrane. The lighter isotope escapes faster than the heavier one under identical experimental conditions. Researchers aim to use this observation to design an isotope separation system based on diffusion principles. Temperature and pressure are maintained constant throughout the experiment. Determine the governing principle behind the observed diffusion difference and evaluate how it can be applied to isotope separation in industrial systems. (5 Marks - [U/P, 3])
28	(i)	In a high-temperature experimental setup, two light nuclei are forced to combine under extreme pressure, producing a heavier nucleus and releasing substantial energy. The system requires continuous input of energy to overcome repulsive forces between nuclei. Scientists are evaluating whether this process can be scaled for long-term energy production. Measurements indicate significantly higher energy output compared to conventional methods. Determine the nature of the reaction and justify the energy behavior by integrating nuclear forces, energy transformations, and feasibility considerations for sustained application. (5 Marks - [U/P, 3])
29	(i)	A medical imaging facility uses a radioactive source that emits highly penetrating radiation with no charge and no mass. The radiation is not deflected in electric or magnetic fields. The nucleus emitting this radiation does not change its atomic or mass number. Researchers aim to interpret this emission type. Identify the radiation type and evaluate its nuclear significance. Interpret why no change occurs in nuclear composition during this emission. (5 Marks - [U/F, 3])
30	(i)	A team of archaeologists recovers wooden artifacts from an ancient settlement buried under sediment layers. They measure the remaining carbon-14 content and compare it with modern atmospheric levels. The decay constant and half-life of carbon-14 are known. The team assumes the organism stopped exchanging carbon after death and that decay followed a predictable exponential pattern. Analyze the process by which the age of the artifact can be determined and justify how the decay principle enables chronological estimation under the given assumptions. (5 Marks - [U/C, 3])
31	(i)	In a nuclear research setup, a radioactive isotope emits fast-moving electrons that show strong deflection in a magnetic field. The emitted radiation has moderate penetration and moderate ionization effects. The nucleus formed after emission shows an increase in atomic number but unchanged mass number. Scientists are evaluating the decay mechanism. Analyze the decay process involved and infer how nuclear composition changes. Reconstruct the reasoning linking particle properties to nuclear transformation. (5 Marks - [U/P, 3])

32	(i)	Atmospheric scientists observe that cosmic rays continuously interact with nitrogen atoms, producing carbon-14. This isotope mixes with atmospheric carbon dioxide and enters plant systems through photosynthesis. Animals then consume these plants, maintaining a dynamic equilibrium of carbon isotopes while alive. Interpret the formation and distribution mechanism of carbon-14 and infer how this continuous cycle supports its use in chronological studies. (5 Marks - [U/P, 3])
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